UNIVERSITAS DIPONEGORO
The Excellent Research University

PROGRAM STUDI D4
TEKNOLOGI REKAYASA OTOMASI
SEKOLAH VOKASI

CAD & CAM I

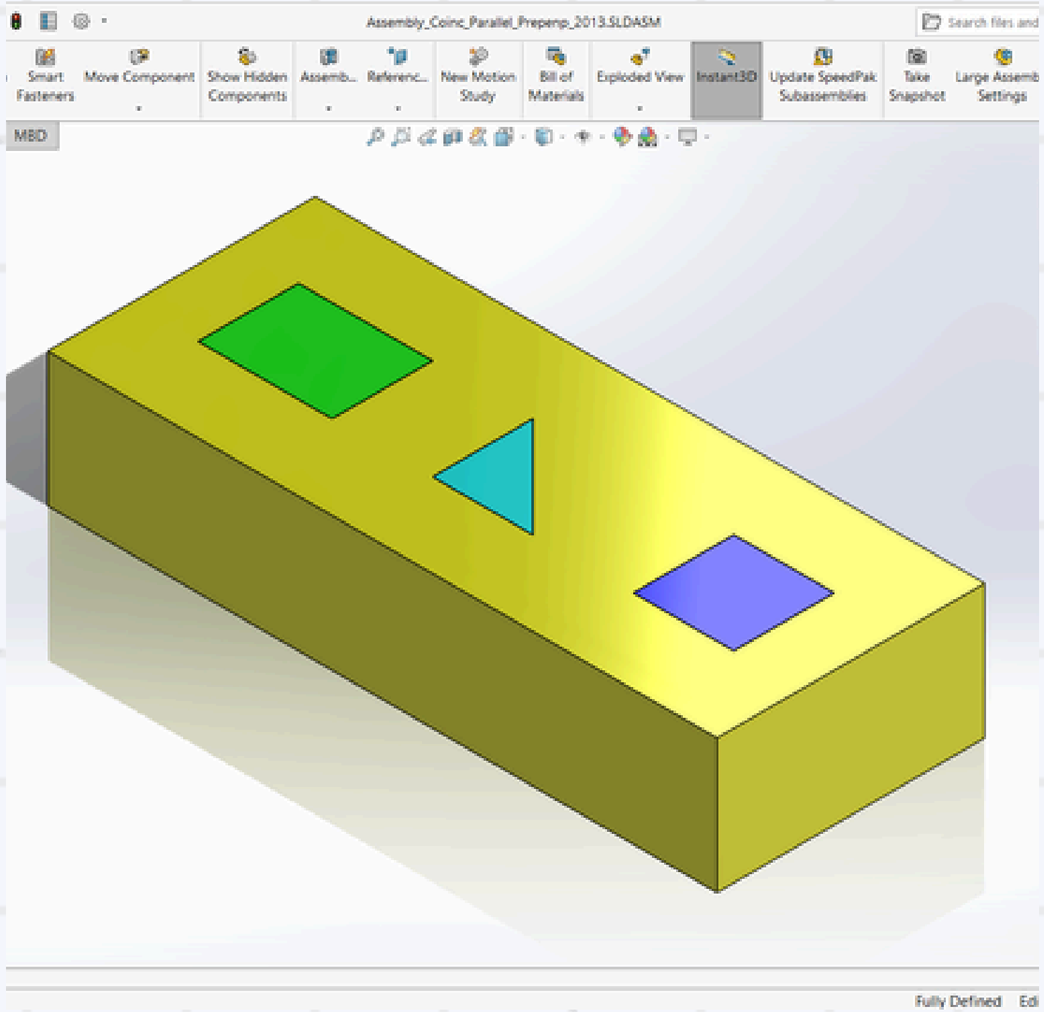
PUTRA WAFI SHAHAN SHAH
4004032562000 6

DOSEN PENGAMPU :

MEGARINI HERSAPUTRI S.T,M.T,
ROFIQ CAHYO PRAYOGO, S.T,M.T.

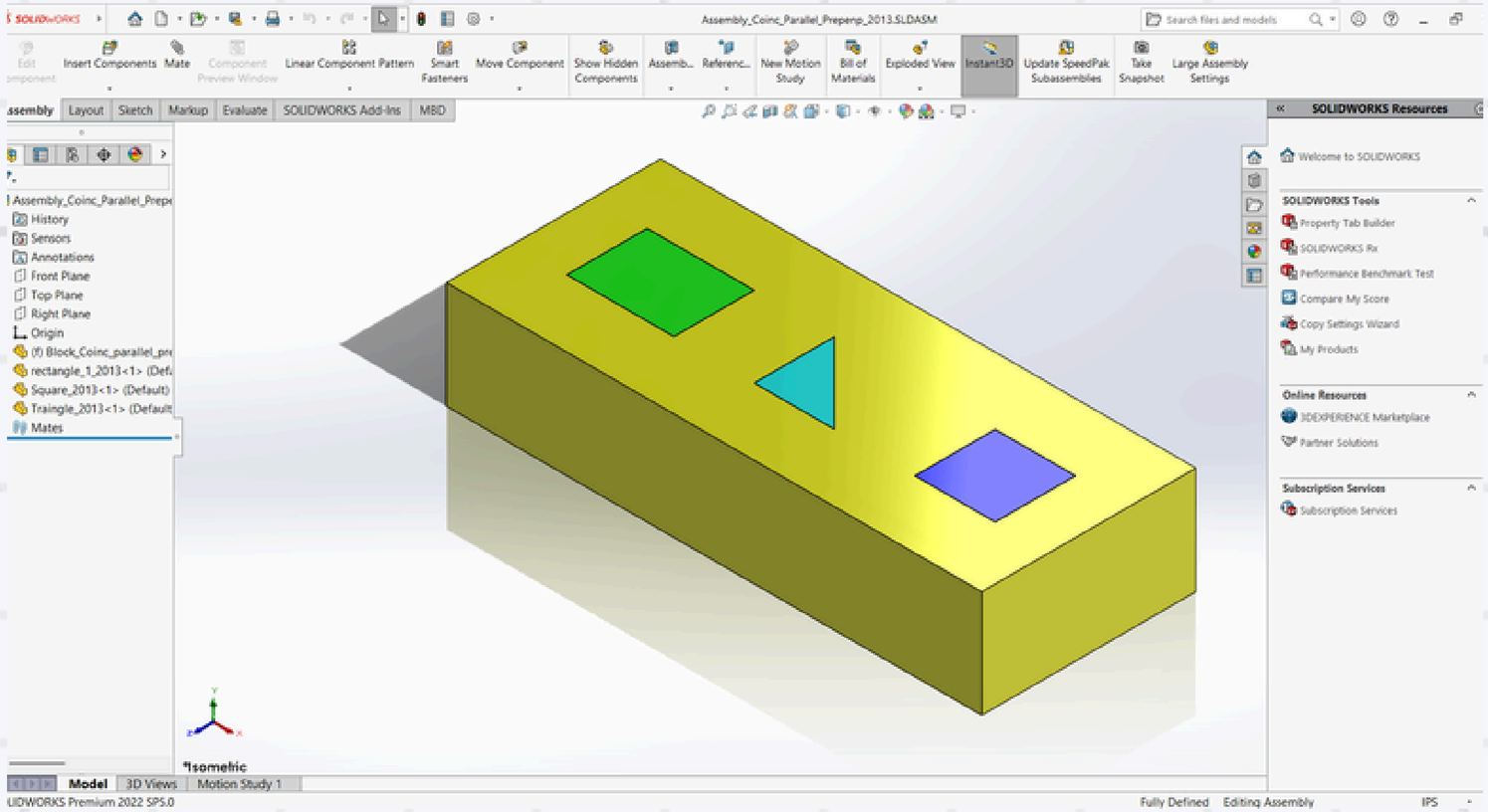
MATERI KALI INI

- CARA MEMBUAT FILE ASSEMBLY
- JENIS-JENIS MATES
- REFERENCE GEOMETRY COORDINATE
- PRAKTEK



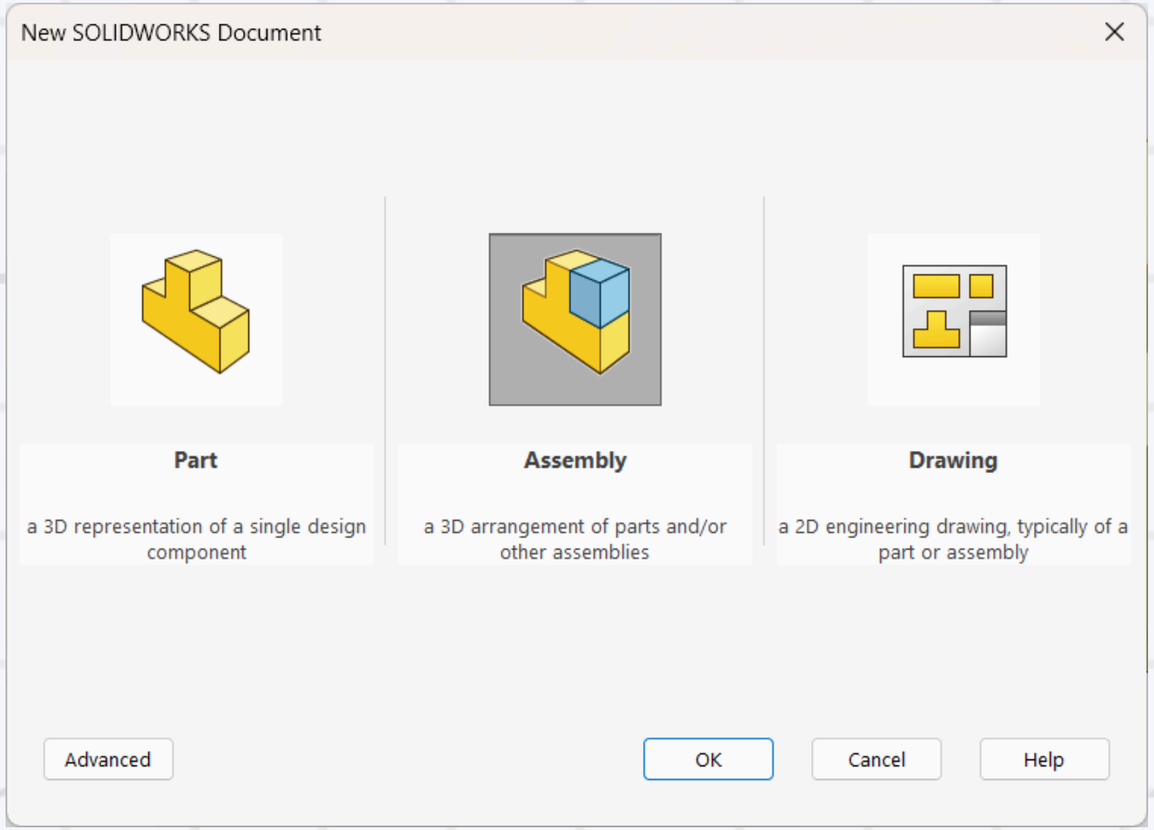
APA ITU ASSEMBLY

Assembly -> Menggabungkan beberapa parts menjadi satu kesatuan yang utuh



CARA MEMBUAT FILE ASSEMBLY

Klik file -> new -> lalu pilih Assembly

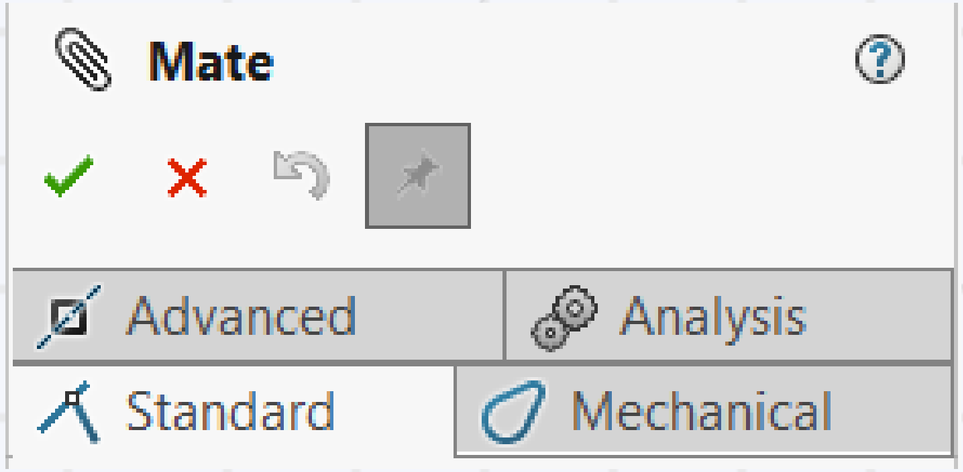


APA ITU MATES

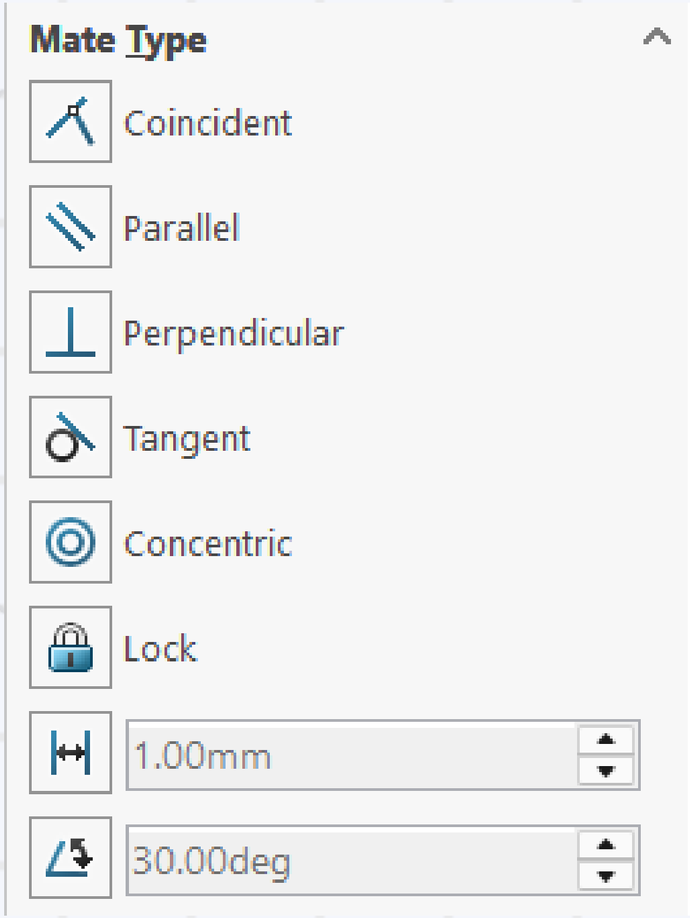
Mates -> fitur relasi/hubungan antar parts supaya saling membentuk satu kesatuan

JENIS-JENIS MATES

- Standar
- Advanced
- Mechanical
- Analysis

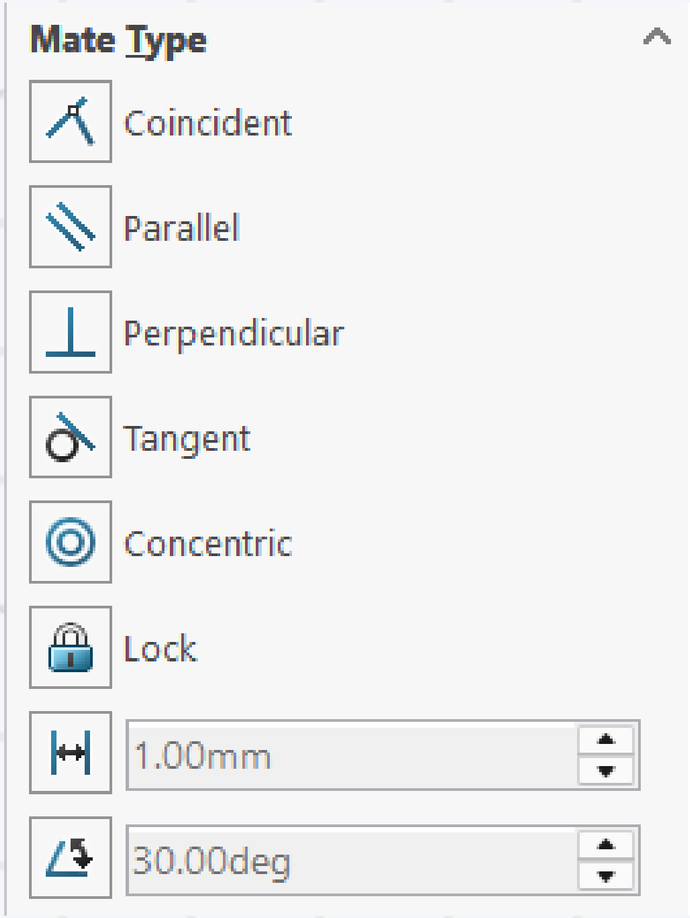


TIPE-TIPE STANDARD MATES



- Coincident = Menggabungkan antara 2 bentuk
- Parallel = Membuat bentuk sejajar
- Perpendicular = Bentuk saling tegak lurus (90°)
- Tangent = Permukaan melengkung menempel pada permukaan lain
- Concentric = Menyamakan titik pusat sesama lingkaran/silinder
- Lock = Mengunci parts supaya tidak berubah-ubah
- Distance = Memberi jarak antara kedua parts
- Angle = Memberi jarak sudut antara kedua parts

TIPE-TIPE STANDARD MATES



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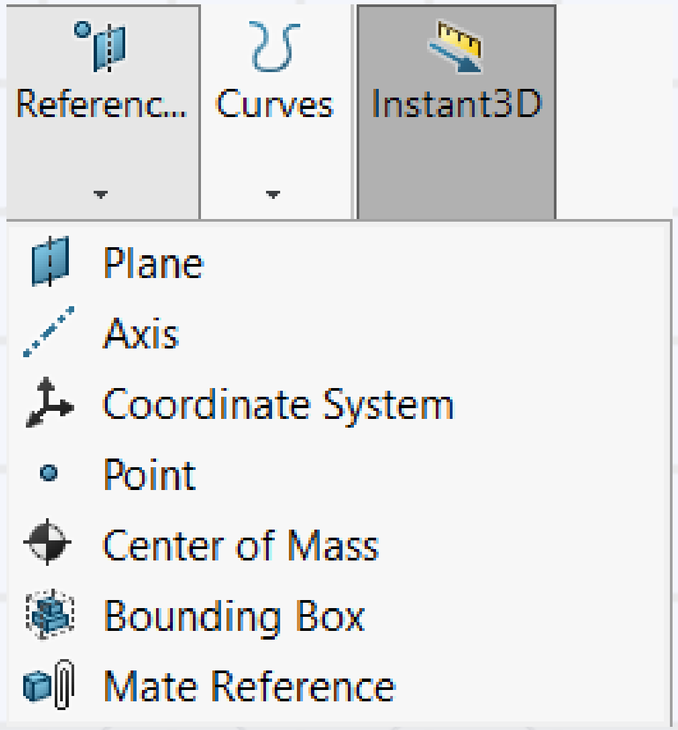
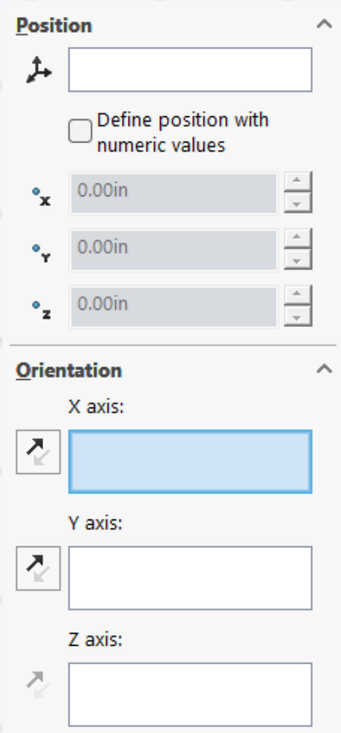
APA ITU REFERENCE GEOMETRY COORDINATE

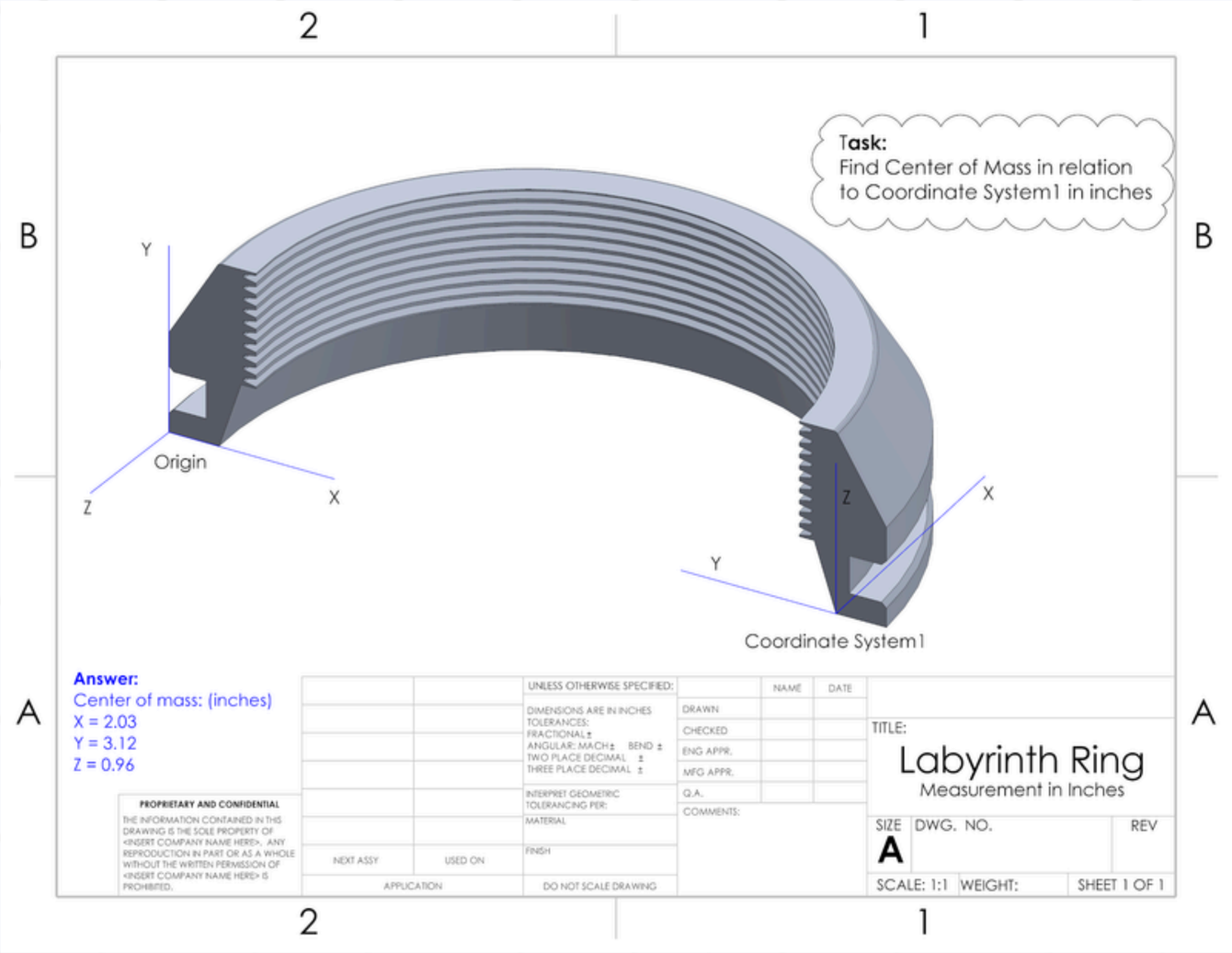
-> Fitur untuk mengetahui titik koordinat tertentu pada komponen

Cara: Klik features -> reference geometry -> lalu pilih Coordinate System

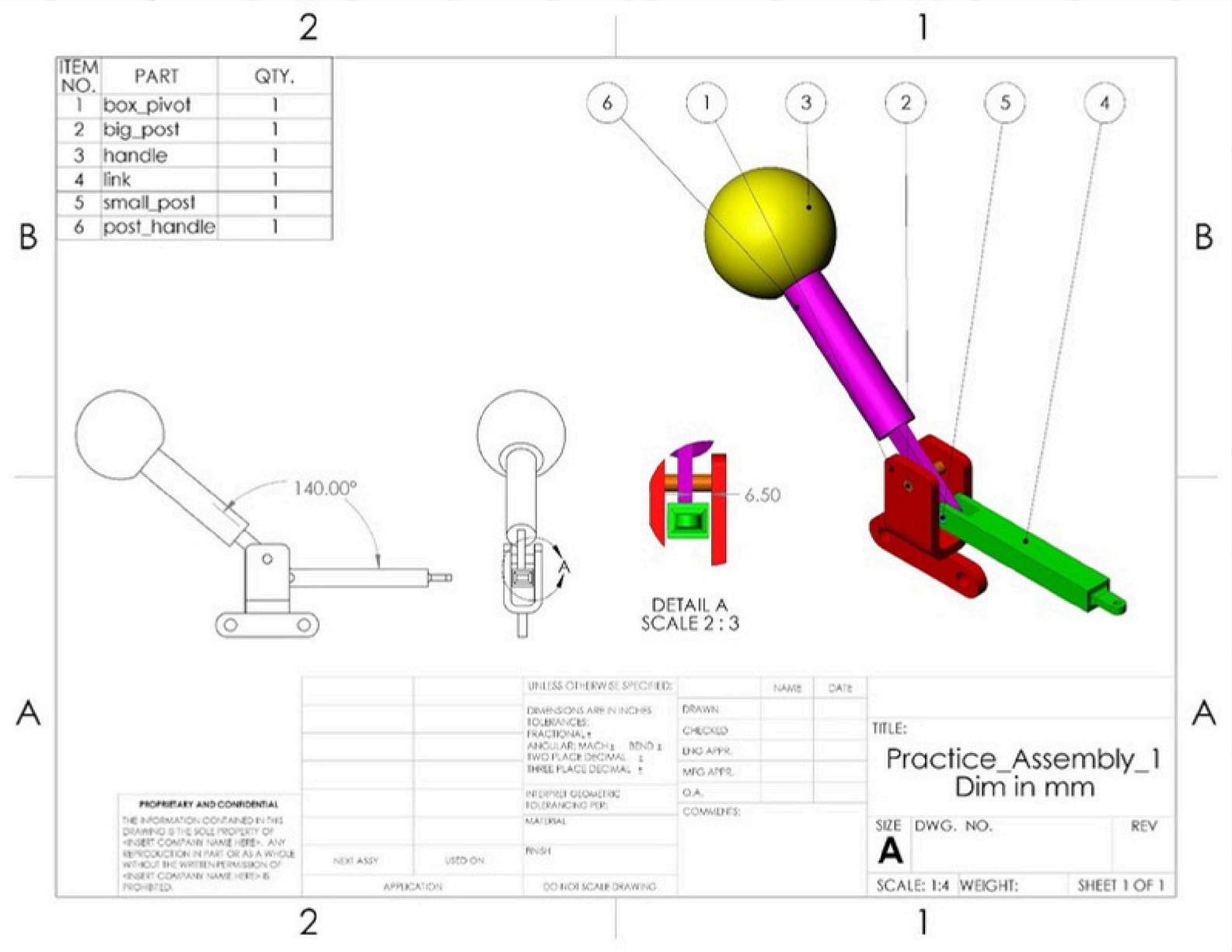




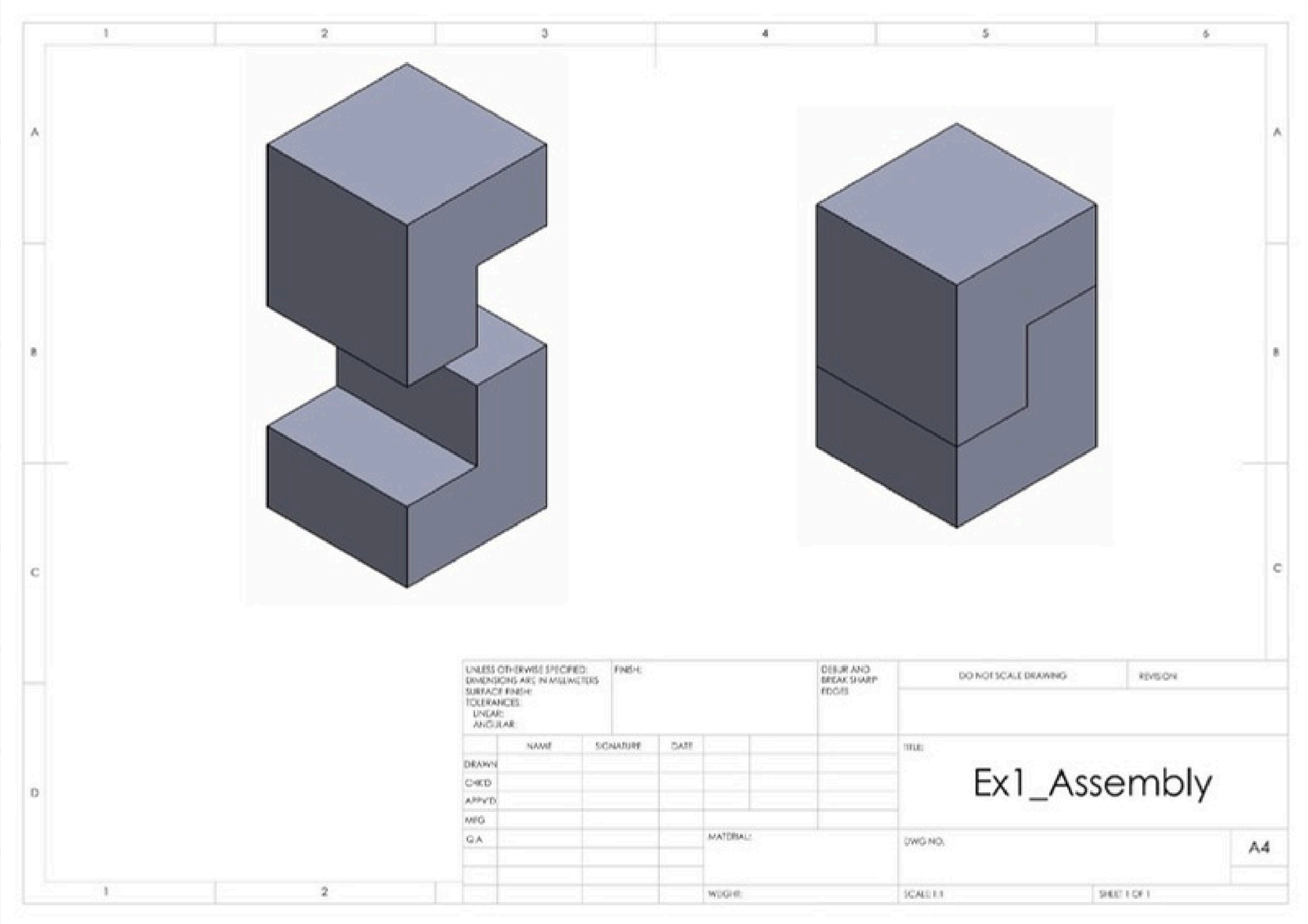
COBA
REFERENCE GEOMETRY COORDINATE



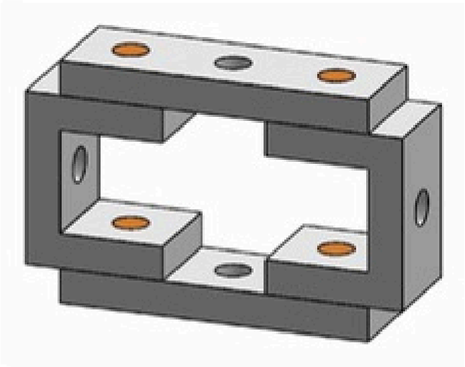
PRAKTEK #I

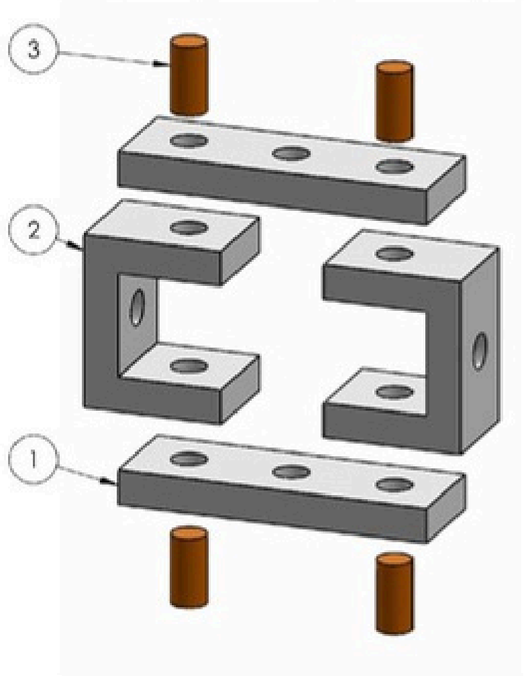

ITEM NO. | PART | QTY.
1 box_pivot 1
2 big_post 1
3 handle 1
4 link 1
5 small_post 1
6 post_handle 1
140.00°
6.50
DETAIL A
SCALE 2 : 3
TITLE:
Practice_Assembly_1
Dim in mm
SIZE A
DWG. NO.
REV
SCALE: 1:4
WEIGHT:
SHEET 1 OF 1
PROPRIETARY AND CONFIDENTIAL
DRAWN
CHECKED
ENG APPR.
MFG APPR.
Q.A.
COMMENTS:
NAME
DATE
NEXT ASSY
USED ON
APPLICATION
DO NOT SCALE DRAWING

PRAKTEK #2


UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:
FINISH:
DEBUR AND BREAK SHARP EDGES
DO NOT SCALE DRAWING
REVISION
NAME
SIGNATURE
DATE
DRAWN
CHK'D
APPV'D
MFG
Q.A.
MATERIAL:
WEIGHT:
TITLE:
Ex1_Assembly
DWG NO.
A4
SCALE: 1:1
SHEET 1 OF 1

PRAKTEK #3





ITEM NO.	PART NUMBER	QTY.
1	Triple Spacer	2
2	C Bracket	2
3	Pin	4

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBUR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHKD			
APPV'D			
MFG			
Q.A			

MATERIAL:

WEIGHT:

TITLE:

EX 2 _ Assembly

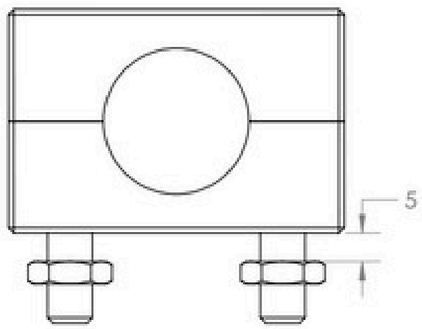
DWG NO.

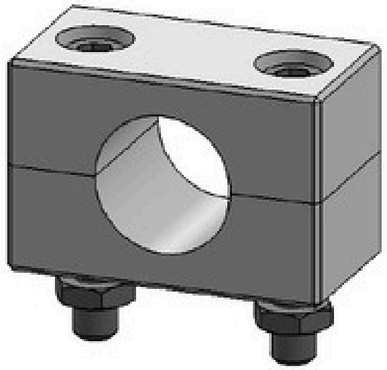
A4

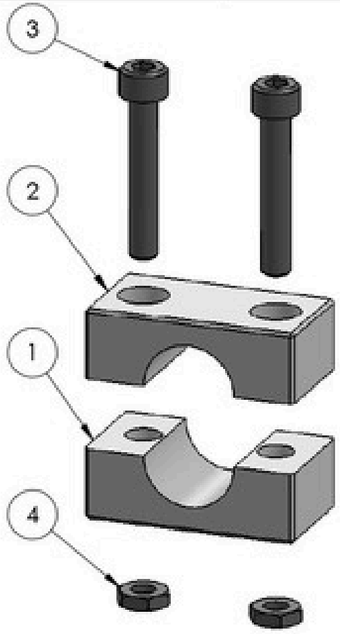
SCALE:1:2

SHEET 1 OF 1

PRAKTEK #4







ITEM NO.	PART NUMBER	QTY.
1	Lower Saddle	1
2	Upper Saddle	1
3	Cap Screw	2
4	Hex Nut	2

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBUR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

MATERIAL:

WEIGHT:

TITLE: EX 3 _ Assembly Dim in MM

DWG NO.

A4

SCALE:1:2

SHEET 1 OF 1

TERIMA KASIH